

Object Detection (Box)

Q: Identify objects to following: <p>Skateboard </p>, <p>Person</p>. Result is text with bounding box coordinate.

Input

Output

Object Detection (Point)

Q: Detect objects from these categories : <p>Hat</p>. Result is text with point coordinate.

Input

Output

OCR

Q: Perform text line-level OCR on the entire image. Result is text with bounding box coordinate.

Input

Output

GUI Grounding

Q: Find and classify objects from the categories: <p>Arab Country Club 1995 mi Arab, AL</p>. Result is text with bounding box coordinate.

Input

Output

Keypoint Detection

Q: Predict the nose, left eye, right eye, left ear, right ear ... for all instances of <p>person</p>.

Input

Output

Referring Detection

Q: Detect all instances of <p>it kind of looks like a flower it's at the bottom of the edifice.</p> in the image.

Input

Output

Visual Prompting

Q: Given visual prompts for <p>object1</p><bbox>[0.527, 0.423, 0.611, 0.538]</bbox>, detect all corresponding instances in the image.

Input

Output

Layout Grounding

Q: Identify objects from the following categories: <p>Header</p>, <p>Title</p>, <p>Figure</p>.

Input

Output

Reasoning Segmentation

Q: <p> what object in the picture can be used to provide natural light and brighten up the room?</p> Please describe the evidence for your segmentation and output the mask.

Input

Output

Referring Segmentation

Q: Could you annotate this image with binary segmentation masks for the categories: <p>a sheep with a black face</p>?"

Input

Output

Interactive Segmentation

Q: provide segmentation masks for this image based on these regions: <point> <input mask>.

Input I

Input II

Output

Generic Segmentation

Q: provide panoptic segmentation masks for this image based on these categories: <p>gravel</p>, <p>sky</p>, <p> tree</p>, <p>window</p>, <p>river</p>, <p>plant flower</p>, <p>sea</p>, <p>umbrella</p>, <p>purse</p>, <p>pedestrian</p>, <p>fruit</p>...

Input

Output I

Output II

GCG Segmentation

Q: Could you please give me a brief description of the image? Please respond with interleaved segmentation masks for the corresponding parts of the answer.

Input

Output I

Output II

Depth

Q: Estimate relative depth for each pixel in the image, with closer objects appearing brighter and distant objects appearing darker.

Input

Output

Surface Normal

Q: Infer the geometric surface orientation for each point in the image.

Input

Output

3D Reconstruction

Q: Using several RGB images of the same scene, estimate the 3D position of each pixel and output XYZ point maps for each image.

Input

Output

Camera Pose Estimation

Q: With the first frame as the reference frame, output the relative pose of all subsequent frames (excluding the first frame).

Input

Output

Structured 2D

Segmentation

Dense Geometry

Multi-view 3D